needs. You add your own rules file in */etc/udev/rules.d/*—the directory meant for local/custom rules.

Create your own rules (which assign the name, symlinks, permissions, etc, that you want) in this directory. To ensure your rule file takes precedence, begin the file name with a number that is lower than the rules file you want to override—for example, *10-local.rules*.

Here are some sample rules with comments, which may help you write your own rules.

Disable root login until the admin connects his USB drive

```
BUS=="usb", SUBSYSTEM=="block", PROGRAM="/bin/enable_root_login"
```

To get this rule to work, you will need to code a program/script with the given name, which obtains the serial numbers of the USB devices connected to the system, and compares each with the serial number of the admin's USB device, which is known. If matched, this program will delete a line from the */etc/pam.d/login* file—"*auth requisite pam_deny.so*" so that root *login* is enabled. There will be no change in the login file if any other USB drive is connected to the system. Conversely, as soon as the USB drive is unplugged, this line will again be added to the login file.

This rule has been checked on RHEL 5.0 and works fine, but when using the *su* command, or when logging in to single-user mode on boot, this rule will not work. To lock down root login for the su command, you can:

1. Edit */etc/security/access.conf* and add *root: ALL*
2. Edit */etc/pam.d/system-auth* and add *account required pam_access.so* as the second line.
3. Edit */etc/pam.d/su* and make *account include system_auth* the first line of this file.

These tasks will, of course, be handled by the *enable_root_login* program. After checking if the attached USB device is the admin's, your program should undo all the changes to the above files, and if the attached device is not the admin's, then apply the changes to the files mentioned above.

This will still not work in single-user mode, so if you need that locked down, perhaps you can assign a GRUB password to prevent easy access to the single-user mode.

To retrieve information like the serial number, kernel name, vendor ID, manufacturer's name, etc, for a particular device, you can run the following command:

```
udevinfo -a -p /sys/block/sdb
```

In newer distributions, *udevinfo* may not be available—in that case, substitute *udevadm info* instead of *udevinfo*.

Disable all USB ports

```
BUS=="usb", OPTIONS+="ignore_device"
```

The effect of this rule will be to disable all devices connected to all USB ports of your system—USB printers, keyboards and mouses will not work. So use it with care!

Disable all USB block devices

```
BUS=="usb", SUBSYSTEM=="block", OPTIONS+="ignore_device"
```

This particular rule prevents USB block storage devices from being recognised. This may be useful for information security or confidentiality within an organisation.

Assign a persistent name to your second IDE disk

```
KERNEL=="hdb", NAME="my_spare"
```

Adjust *hdb* if you want to apply this rule to another drive. It could be *sdb* for SCSI disks, for example.

Ignore the second USB SCSI/IDE disk connected via USB

```
BUS=="usb", KERNEL=="sdb", OPTIONS+="ignore_device"
```

OR

```
BUS=="usb", KERNEL=="hdb", OPTIONS+="ignore_device"
```

Add a symbolic link to a specific USB mouse device's name

```
SUBSYSTEM=="input", BUS=="usb", SYSFS{serial}=="0000:00:1d.0",
SYMLINK+=="MY-USB-MOUSE"
```

Change device node name based on the manufacturer's name

```
BUS=="usb", SYSFS{manufacturer}=="JetFlash", NAME="UNIVERSE"
```

This rule states that if the name of a USB drive's manufacturer is JetFlash, the device's node name will be 'UNIVERSE'.

Selectively allow USB block devices via a custom *program*

```
BUS=="usb", SUBSYSTEM=="block", PROGRAM="/bin/usbc.jar", RESULT!="my",
OPTIONS+="ignore_device"
```

If the output generated by the program is equal to the 'my' value that is specified, then allow the USB device; if the output is anything else, then ignore the device. For further reference, visit:

- http://www.kernel.org/pub/linux/utils/kernel/hotplug/udev.html
- http://reactivated.net/writing_udev_rules.html

(* All information provided in this article has been tested on RHEL 4.0 and higher) END

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